light is measured which turns out to be exactly in accordance with the prediction.

But it is not only space that curves: time does, too. Einstein predicts that time on Earth passes more quickly at higher altitude, and more slowly at lower altitude. This is measured, and also proves to be the case. Today we have extremely precise clocks, in many laboratories, and it is possible to measure this strange effect even for a difference in altitude of just a few centimetres. Place a watch on the floor and another on a table: the one on the floor registers less passing of time than the one on the table. Why? Because time is not universal and fixed, it is something which expands and shrinks, according to the vicinity of masses: the Earth, like all masses, distorts spacetime, slowing time down in its vicinity. Only slightly – but two twins who have lived respectively at sea-level and in the mountains will find that, when they meet up again, one will have aged more than the other (figure 3.8).

This effect offers an interesting explanation as to why things fall. If you look at a map of the world and the route taken by an aeroplane flying from Rome to New York, it does not seem to be straight: the aeroplane makes an arc towards the north. Why? Because, the Earth being curved, crossing northwards is shorter than keeping to the same parallel. The distances between meridians are shorter the more northerly you are; therefore, it is better to head northwards, to shorten the route (figure 3.9).

Well, believe it or not, a ball thrown upwards falls downwards for the same reason: it ‘gains time’ moving higher up, because time passes at a different speed up there. In both cases, aeroplane and ball follow a straight trajectory in a space (or spacetime) that is curved (figure 3.10).[fn12]